DCU

# DUBLIN CITY UNIVERSITY

## SEMESTER ONE RESIT EXAMINATIONS 2012

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| **MODULE:** **(Title & Code)** | CA320 Computability & Complexity |
| **COURSE:** | BSc. in Computer Applications (SE Stream) |
| **YEAR:** | 3 |
| **EXAMINERS:** **(Including Telephone Nos.)** | Dr. J. Power,<br>Dr. M. Manzke,<br>Dr. D. Sinclair, Ext no. 5510. |
| **TIME ALLOWED:** | 2 hours |
| **INSTRUCTIONS:** | Answer 4 Questions.<br>At least 1 question must be from Section A.<br>All questions carry equal marks. |

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones

# Section A

## QUESTION 1 [TOTAL MARKS: 25]

**1(a)** **[5 Marks]**

What are function types in Haskell? How is a function type written in Haskell? What is the implication of the particular form Haskell uses for defining a function type and invoking functions?

**1(b)** **[5 Marks]**

What are *class constraints* in Haskell? Give an example of how to use a *class constraint*.

**1(c)** **[15 Marks]**

Write a Haskell function, **without using the reverse function** , that tests whether or not a list is a "palindrome", i.e. the list is equal to its reverse. The function should use class constraints to ensure that it will work for lists of integers, floats, and characters.

[End of Question 1]

## QUESTION 2 [TOTAL MARKS: 25]

**2(a)** **[5 Marks]**

Describe, with the use of examples, the *case expression* in Haskell.

**2(b)** **[5 Marks]**

In the context of Haskell, what is meant by *list comprehension* and give an example of it.

**2(c)** **[15 Marks]**

Implement the *Quicksort* algorithm in Haskell.

[End of Question 2]

# Section B

**QUESTION 3** **[TOTAL MARKS: 25]**

**3(a)** **[5 Marks]**

Briefly describe the difference between *computability* and *complexity*.

**3(b)** **[5 Marks]**

What is a *pumping lemma*?

**3(c)** **[15 Marks]**

Compare and contrast the following:

- Deterministic Finite Automata
- Pushdown Automata
- Linear Bounded Automata

[End of Question 3]

**QUESTION 4** **[TOTAL MARKS: 25]**

**4(a)** **[10 Marks]**

Prove that the set $\mathcal{Q} = \{\frac{m}{n} | m, n \in \mathcal{N}\}$, the set of positive rational numbers, countable.

**4(b)** **[10 Marks]**

Prove that the *Halting Problem* is undecidable.

**4(c)** **[5 Marks]**

Give two examples of other undecidable problems.

[End of Question 4]

**QUESTION 5** **[TOTAL MARKS: 25]**

**5(a)** **[5 Marks]**

Describe the complexity classes **P** and **NP** and what is the relationship between them?

5(b) [10 Marks]

Prove that a language $L$ belongs to **NP** if and only if a language $L_{check} \in \mathbf{P}$ and a polynomial $p$ exist such that

$$L = \{x | \exists y.x, y \in L_{check} \wedge y \leq p(x)\}$$

5(c) [10 Marks]

Describe a problem that belongs to the class **NP** and, using the theorem in part(b) of this question, prove that it is a **NP** problem.

[End of Question 5]

[END OF EXAM]